DIGITAL SAT & EST MATH COURSES

# Rabaa Farrag

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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

# EST1 March 2026

## Official Exam • Full Practice Test

Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

| 50 | MCQ | 75 min |
|---|---|---|
| QUESTIONS | MULTIPLE CHOICE | TIME LIMIT |

NAME _____ DATE _____ SCORE _____ /50

### How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.


RF

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**Eng. Rabaa Farrag**

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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# Practice Questions

50 QUESTIONS

**1** **Question 1.** An amount of $14,000 is divided among three sisters, Layla, Yasmine, and Aya, in the ratio $1 : 2 : 4$, respectively. How much did Yasmine receive?

**A.** $2,000
**B.** $4,000
**C.** $7,000
**D.** $8,000

**2** What is the median of the set of data above?

$$6, 7, 7, 6, 1, 5, 9, 9, 7, 10$$

(A) 5.0
(B) 6.5
(C) 7.0
(D) 7.5

**3** **Questions 3 – 5** refer to the information below:

| Brand | Number of cars rented | Revenue earned (in $) |
|---|---|---|
| Hyundai | 14 | 134,400 |
| Toyota | 8 | 85,440 |
| Jeep | 19 | 239,400 |
| Honda | 13 | 117,000 |

A car rental company leased a total of 54 cars of various brands to an embassy for a period of one year. The table above shows the numbers of cars rented for each brand along with the total revenue earned at the end of the year.

**Question 3.** What is the monthly rental cost paid by the embassy for each Hyundai car?

**A.** $750
**B.** $760
**C.** $775
**D.** $800

**4** **Questions 3 – 5** refer to the information below:

| Brand | Number of cars rented | Revenue earned (in $) |
|---|---|---|
| Hyundai | 14 | 134,400 |
| Toyota | 8 | 85,440 |
| Jeep | 19 | 239,400 |
| Honda | 13 | 117,000 |

**Question 4.** At the end of the year, the company paid 11% of its total revenue to the government as taxes. What was the total amount paid as taxes for the year?

**A.** $14,784.0
**B.** $37,052.4
**C.** $50,770.6
**D.** $63,386.4

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**5** Questions 5 – 7 refer to the information below.

| Brand | Number of cars rented | Revenue earned (in $) |
|---|---|---|
| Hyundai | 14 | 134,400 |
| Toyota | 8 | 85,440 |
| Jeep | 19 | 239,400 |
| Honda | 13 | 117,000 |

**Question 5.** The company offered the embassy a 20% discount on all Jeep rentals as a gesture of appreciation for renewing the contract for an additional year. What will be the new monthly rental cost per Jeep for the second year?

**A.** $800
**B.** $840
**C.** $850
**D.** $900

**6** The probability that Mohammad successfully scores a penalty in a football match is 0.3. If he takes 3 penalty shots in a single game, what is the probability that he scores all three?

(A) 0.027
(B) 0.343
(C) 0.540
(D) 0.90

**7** How many possible arrangements can be formed using all the letters in the word OFFICE?

(A) 180
(B) 360
(C) 720
(D) 1,440

**8** **Question 8.** What is the equation of the line passing through $A(-9, 3)$ and $B(1, 4)$?

**A.** $y = -x + 5$
**B.** $y = 0.1x - 3.9$
**C.** $y = 0.1x + 3.9$
**D.** $y = x + 3$

**9** **Question 9.** Given $f(x) = -3(x + 4) - 4x^2$ and $g(x) = 2x^2 - x + 1$, the value of $\frac{f(1)}{g(1)}$ is:

**A.** −9.5
**B.** −4.0
**C.** 4.5
**D.** 9.5

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**12** The price of a mobile phone increased by 40%, resulting in a final price of $308. What was the original price of the mobile phone before the increase?

A) $220
B) $250
C) $280
D) $288

**13** Which of the following is not a factor of $P(x) = 3x^4 + 5x^3 - 64x^2 - 164x - 80$?

A) $x - 5$
B) $x - 2$
C) $x + 4$
D) $3x + 2$

**14** Jalal invested $4,500 for 2 years at an annual simple interest rate of 2%. How much interest did he earn at the end of the 2 years?

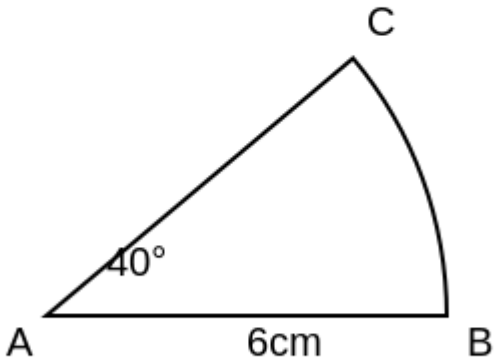
A) $90.0
B) $180.0
C) $181.8
D) $300.2

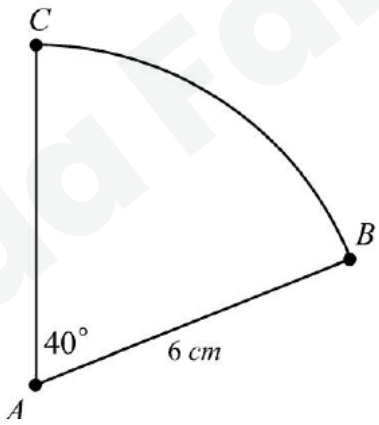
**15** If $3x - 7 = 20$, what is the value of $2x$?

A) 4.5
B) 9.0
C) 12.5
D) 18.0

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**16** What is the approximate perimeter of the shape shown in the figure above? (Figure not drawn to scale).




(A) 10.19 cm
(B) 12.0 cm
(C) 14.73 cm
(D) 16.19 cm

**17** A book contains 400 pages, of which 4 are defective. If 230 copies of the book were distributed each week over the past 5 weeks, how many non-defective pages were distributed in total?

(A) 4,600
(B) 91,080
(C) 182,160
(D) 455,400

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**18** Which of the following is a simplified form of $\frac{\sqrt{125a^2}}{\sqrt{5y^3}}$?

A $\frac{|a|\sqrt{5y}}{y^2}$

B $\frac{5|a|\sqrt{y}}{y^2}$

C $\frac{5|a|\sqrt{5y}}{y^2}$

D $5|a|\sqrt{5y}$

**19** Which of the following is a solution of $|3x - 5| + 1 = 2$?

A $1/3$

B $1$

C $2$

D $7/3$

**20** Which of the following is the equation of the axis of symmetry of the graph of $f(x) = 4x^2 - 20x + 7$?

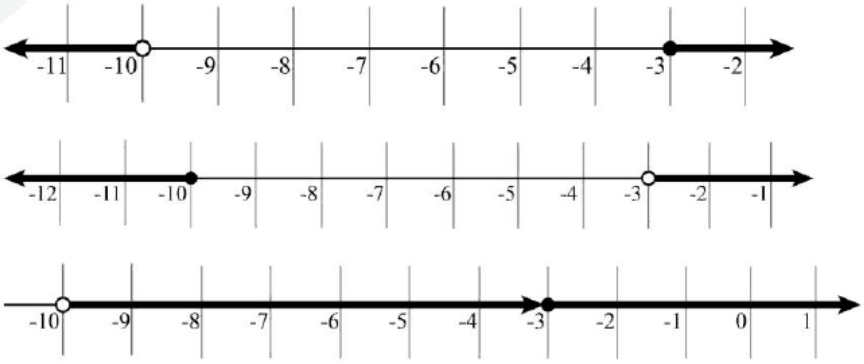
A $y = -18$

B $x = -2.5$

C $x = 1.5$

D $x = 2.5$

**21** Which of the following is a solution of $|3x - 5| + 1 = 2$?




A $1/3$

B $1$

C $2$

D $7/3$

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**23** Which of the following has infinitely many solutions?

(A) $4x - 1 = 9$

(B) $|3x + 7| < 4$

(C) $2x - 6 = 3(x + 7)$

(D) $|-4x + 1| > -5$

**24** **Questions 24 and 25** refer to the information below:

Consider the function $f(x) = 3ax^3 + 4bx^2 + x + x^3$, where $a$ and $b$ are two real numbers.

**Question 24.** In order to make the equation quadratic, $a$ must be equal to:

**A.** $-1$

**B.** $-\frac{1}{3}$

**C.** $0$

**D.** $\frac{1}{2}$

**25**

**D.** $\frac{1}{3}$

**Question 25.** If $a = 1$, and the coefficient of the $x^2$ term is equal to the coefficient of the $x^3$ term, then what is the value of $b$?

**A.** $-1$

**B.** $\frac{1}{2}$

**C.** $1$

**D.** $\frac{3}{2}$

**26** In the table above, x and y represent a linear relationship. What is the value of 2a?

| $x$ | $y$ |
|---|---|
| 2 | 5 |
| 5 | $a$ |
| 7 | 15 |

(A) 5.5

(B) 13.0

(C) 17.5

(D) 22.0

**27** If $1.2\ m^3$ of a pool can be filled by 9 gallons of water, how many gallons are needed to fill a $26\ m^3$ pool?

(A) 195

(B) 202

(C) 223

(D) 234

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**31** The number of players in a basketball academy increased from 20 to 34 in one month. If m% is the percent increase, then m is equal to:

(A) 30
(B) 45
(C) 50
(D) 70

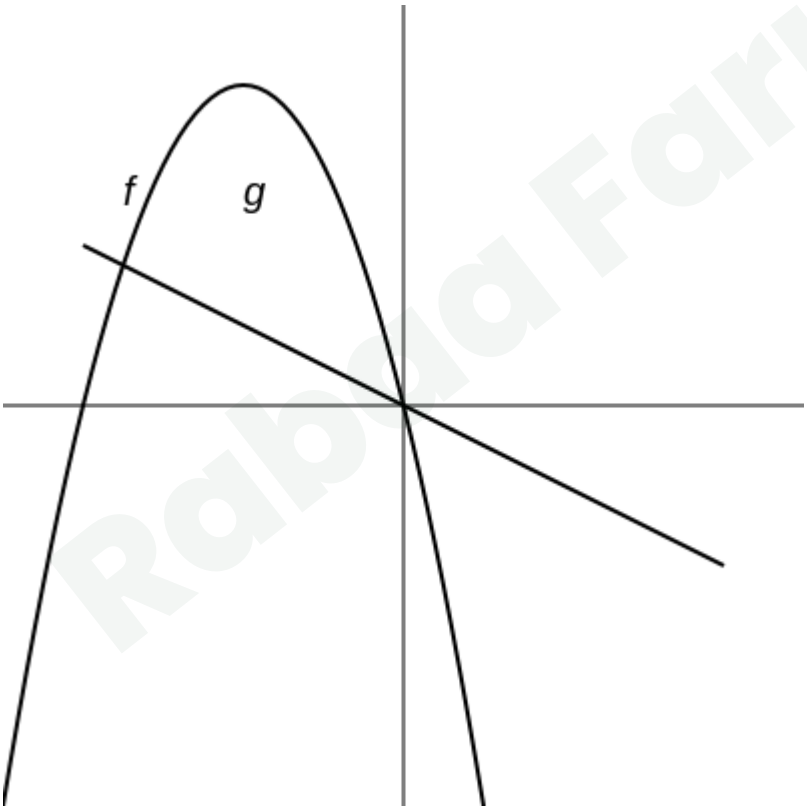
**32** If $f(x) = 2x + 1$ and $g(x) = -3x + 1$, then what is the value of $(f + g)(1)$?

(A) 1
(B) 3
(C) 6
(D) 7

**33** If $3x - 7 = 20$, what is the value of $2x$?

(A) 4.5
(B) 9.0
(C) 12.5
(D) 18.0

**34** What is the value of $f(-6) + g(-1)$?




(A) 1
(B) 3
(C) 4
(D) 8

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**35** What is the slope of the line passing through $A(-3, 4)$ and $B(1, -2)$?

(A) -1.5
(B) 0.5
(C) 1.0
(D) 1.5

**36** Rounding $0.05674$ kilograms to the nearest gram is equivalent to:

(A) 6 g
(B) 56 g
(C) 57 g
(D) 567 g

**37** What is the discriminant of the equation $3x^2 - 3x + 1 = 0$?

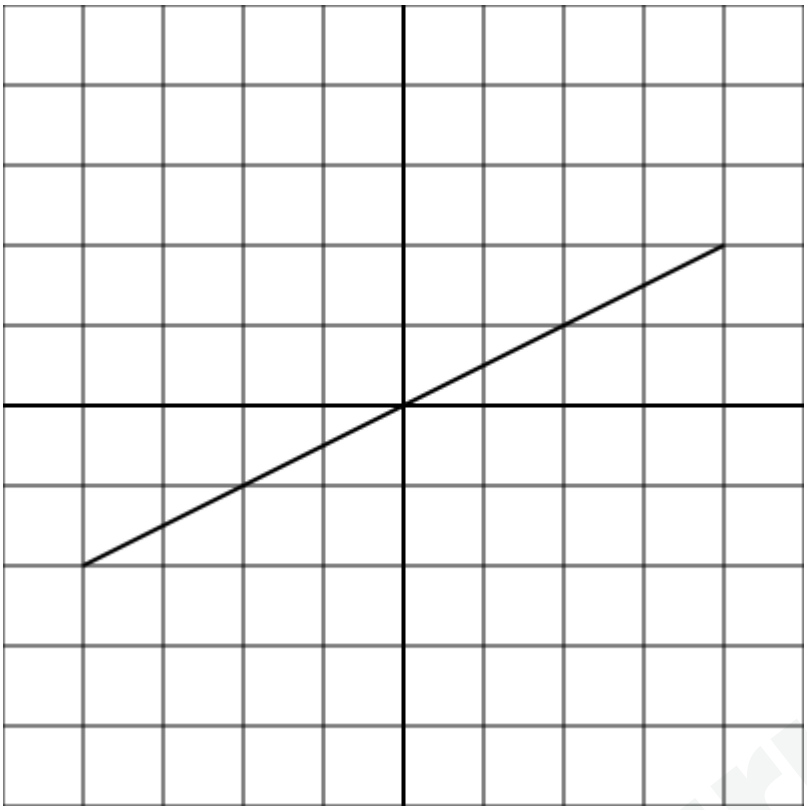
(A) 3
(B) -3
(C) -15
(D) -21

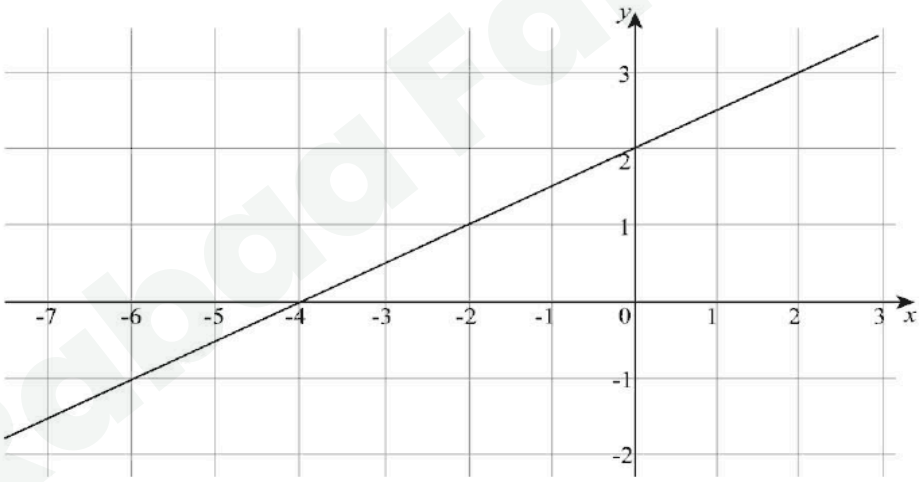
**38** If $25^{3x-1} = 125^{x}$, then $x =$ ?

(A) $1/3$
(B) 0
(C) $2/3$
(D) 1

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**39** Which of the following is the equation of the line perpendicular to the line graphed above and passing through $(1, 0)$?






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A $y = -2x + 2$

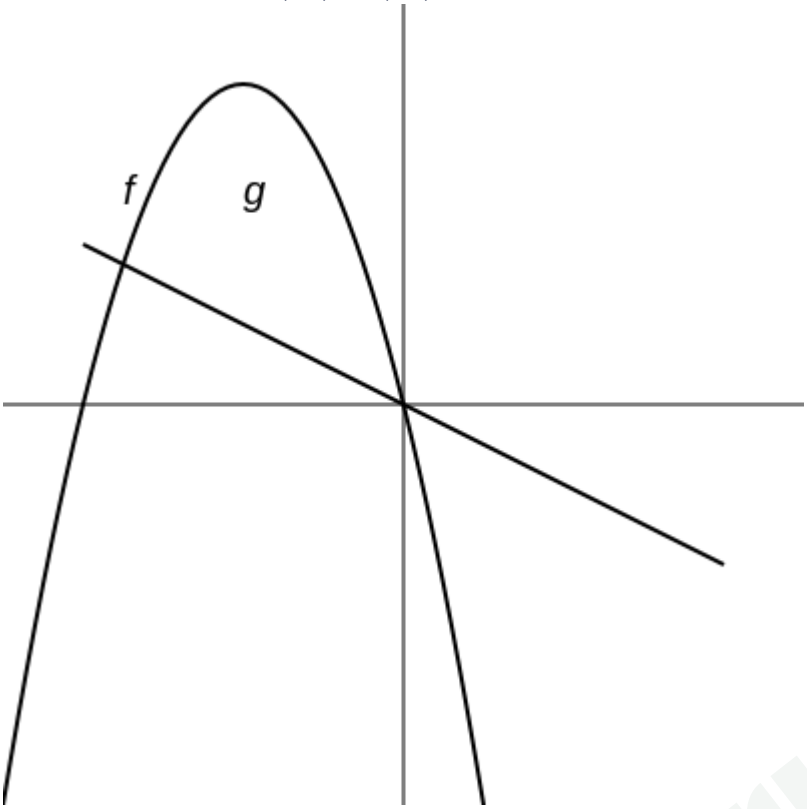
B $y = -2x + 1$

C $y = \frac{1}{2}x - \frac{1}{2}$

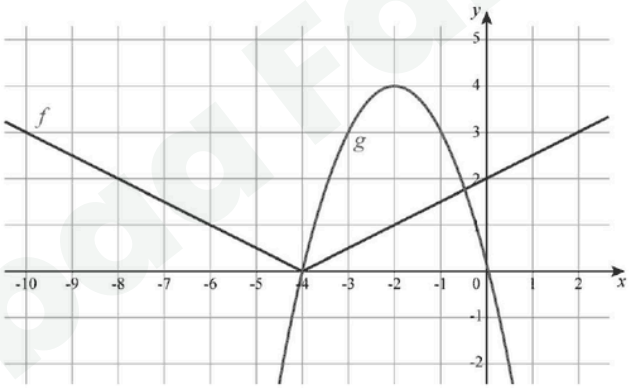
D $y = 2x - 2$

**40** The coordinates of the highest point in the graph of the function $f(x) = -2x^2 - 4x + 1$ are:

**42** What is the value of $f(-6) + g(-1)$?



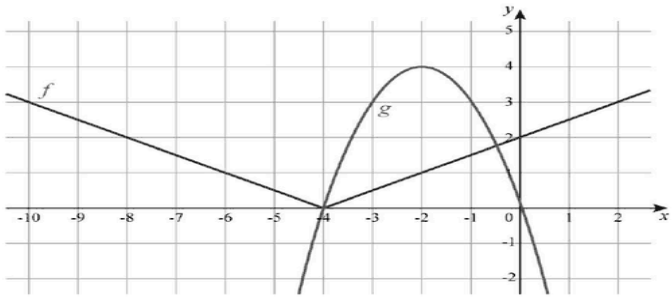

**Questions 42 and 43** refer to the graph below:




(A) 1
(B) 3
(C) 4
(D) 8

**43**

**Questions 42 and 43** refer to the graph below:




**Question 43.** Which of the following represents the solution set of $g(x) > 3$?

**A.** $[-3, -1]$
**B.** $(-\infty, -3) \cup (-1, +\infty)$
**C.** $(-\infty, -1) \cup (-1, +\infty)$

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**45** Solving $2x + 3yx + 1 = 4x^2$ for $y$ will give:

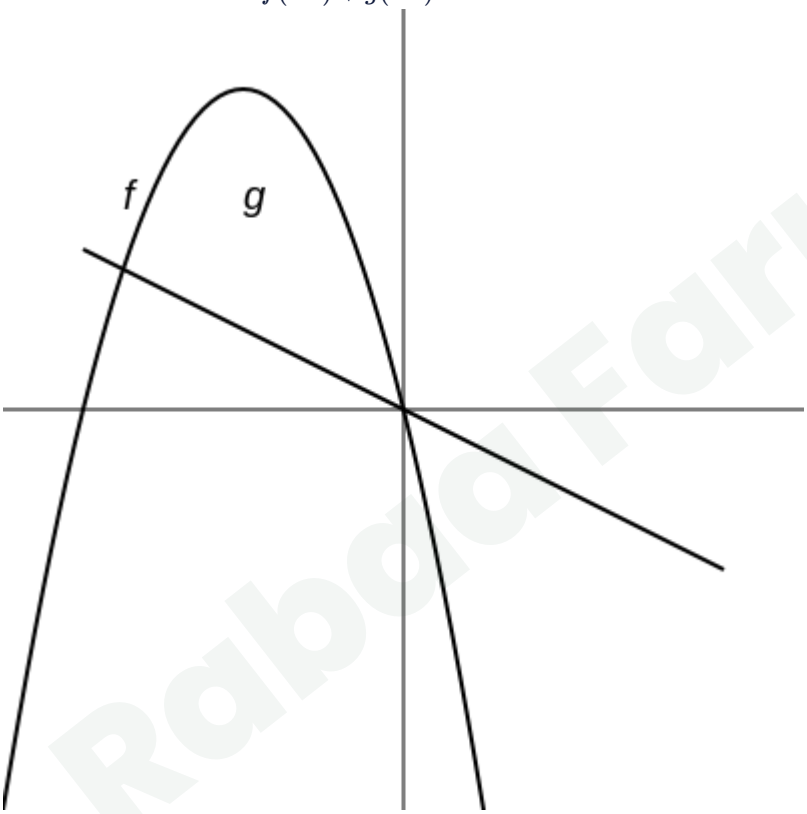
(A) $y = \dfrac{4x^2 - 2x - 1}{3x}$

(B) $y = 4x^2 - 2x - 1$

(C) $y = 4x^2 - 5x - 1$

(D) $y = \dfrac{4x^2 + 2x + 1}{3x}$

**46** What is the value of $f(-6) + g(-1)$?




(A) 1

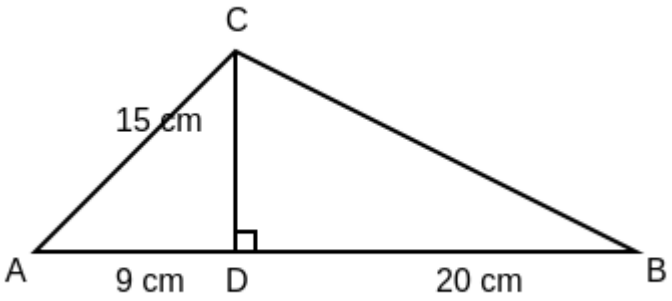
(B) 3

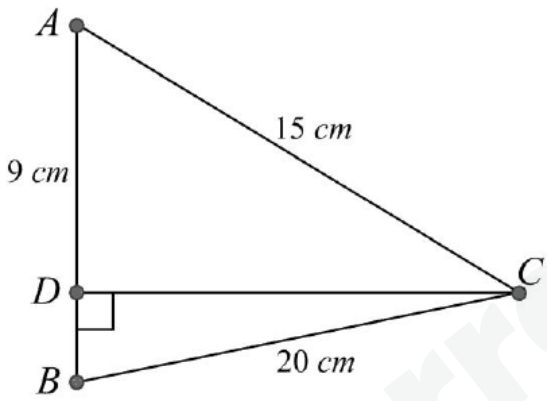
(C) 4

(D) 8

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**47** The area of $\Delta ABC$ in the figure above is: (Figure not drawn to scale)




A) 131.5 cm^2
B) 144.5 cm^2
C) 150.0 cm^2
D) 300.0 cm^2

**48** If $3x - 7 = 20$, what is the value of $2x$?

A) 4.5
B) 9.0
C) 12.5
D) 18.0

**49** If $P(x) = 3x^3 - 2x + m$ is divisible by $(x + 1)$, then the value of $m$ is:

A) 1
B) 3
C) 6
D) 9

**50** What is the product of the positive integers in the solution set of $3x + 1 < 10$?

A) 1
B) 2
C) 3
D) 6

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# Answer Key

50 ANSWERS

| Q | ANSWER | Q | ANSWER |
|---|---|---|---|
| 1 | B | 26 | D |
| 2 | C | 27 | A |
| 3 | D | 28 | C |
| 4 | D | 29 | C |
| 5 | B | 30 | A |
| 6 | A | 31 | D |
| 7 | B | 32 | A |
| 8 | C | 33 | D |
| 9 | A | 34 | C |
| 10 | C | 35 | A |
| 11 | C | 36 | C |
| 12 | A | 37 | B |
| 13 | B | 38 | C |
| 14 | C | 39 | A |
| 15 | C | 40 | C |
| 16 | D | 41 | B |
| 17 | D | 42 | C |
| 18 | B | 43 | D |
| 19 | C | 44 | A |
| 20 | D | 45 | A |
| 21 | D | 46 | A |
| 22 | A | 47 | C |
| 23 | D | 48 | C |
| 24 | B | 49 | A |
| 25 | C | 50 | B |

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# Detailed Solutions

50 SOLUTIONS

**1** **Answer: B**

See the answer key.

**2** **Answer: C** 7.0

Ordered data: 1, 5, 6, 6, 7, 7, 7, 9, 9, 10. There are 10 values, so median is the average of the 5th and 6th values: (7+7)/2 = 7.0.

**3** **Answer: D**

See the answer key.

**4** **Answer: D**

See the answer key.

**5** **Answer: B**

See the answer key.

**6** **Answer: A** 0.027

Probability = 0.3 * 0.3 * 0.3 = 0.027.

**7** **Answer: B** 360

The word OFFICE has 6 letters with 'F' repeating 2 times. Total arrangements = 6! / 2! = 720 / 2 = 360.

**8** **Answer: C**

See the answer key.

**9** **Answer: A**

See the answer key.

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**10** **Answer: C**

See the answer key.

**11** **Answer: C**

See the answer key.

**12** **Answer: A** $220

Let x be original price. x * 1.40 = 308. x = 308 / 1.40 = 220.

**13** **Answer: B** $x - 2$

Using Remainder Theorem, test roots. For x=2: P(2) = 3(16)+5(8)-64(4)-164(2)-80 = 48+40-256-328-80 = -576 != 0. Thus x-2 is not a factor.

**14** **Answer: C** $181.8

Simple Interest I = P * r * t = 4500 * 0.02 * 2 = 180.

**15** **Answer: C** 12.5

$3x = 27 \Rightarrow x = 9$. Thus $2x = 18$.

**16** **Answer: D** 16.19 cm

Perimeter = 2 * r + arc_length. r = 6, angle = 40 degrees. arc = 6 * (40*pi/180) = 4.188. Perimeter = 6 + 6 + 4.188 = 16.19.

**17** **Answer: D** 455,400

Non-defective per book = 396. Total books = 230 * 5 = 1150. Total non-defective = 1150 * 396 = 455,400.

**18** **Answer: B** $\frac{5|a|\sqrt{y}}{y^2}$

Fraction = $\sqrt{25a^2/y^3} = 5|a|/y\sqrt{y}$. Multiply by $\sqrt{y}/\sqrt{y}$ to rationalize denominator: $5|a|\sqrt{y}/y^2$.

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**19** **Answer: C** 2

$|3x - 5| = 1$ gives $3x - 5 = 1 \Rightarrow 3x = 6 \Rightarrow x = 2$.

**20** **Answer: D** $x = 2.5$

$x = -b/2a = -(-20)/(2 * 4) = 20/8 = 2.5$.

**21** **Answer: D** $7/3$

$|3x - 5| = 1$ gives $3x - 5 = 1 \Rightarrow 3x = 6 \Rightarrow x = 2$.

**22** **Answer: A** 3

Midpoint $y = (y_1 + y_2)/2 = (-5 + 11)/2 = 6/2 = 3$.

**23** **Answer: D** $|-4x + 1| > -5$

Absolute value is always non-negative, so it is always strictly greater than a negative number like -5. Therefore, $|-4x + 1| > -5$ is true for all real numbers.

**24** **Answer: B**

See the answer key.

**25** **Answer: C**

See the answer key.

**26** **Answer: D** 22.0

Slope $m = (15 - 5)/(7 - 2) = 10/5 = 2$. Using $x = 5$: $y - 5 = 2(5 - 2) \Rightarrow a - 5 = 6 \Rightarrow a = 11$. Then $2a = 22$.

**27** **Answer: A** 195

Set up proportion: $9/1.2 = x/26 \Rightarrow x = (9 \times 26)/1.2 = 234/1.2 = 195$.

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**28** **Answer: C** 2,304

$V = (4/3)\pi r^3 = (4/3)\pi(12)^3 = (4/3)\pi(1728) = 2304\pi$. Thus $k = 2304$.

**29** **Answer: C** 4

Testing factors of 24. For $x = -4$, $(-4)^3 - (-4)^2 - 14(-4) + 24 = -64 - 16 + 56 + 24 = 0$. So $-4$ is a root. Its absolute value is 4.

**30** **Answer: A** 1

$3x \leq 5 \Rightarrow x \leq 5/3 \approx 1.67$. The greatest positive integer is 1.

**31** **Answer: D** 70

Increase = 34 - 20 = 14. Percent increase = (14 / 20) * 100 = 70%.

**32** **Answer: A** 1

$f(1) = 2(1) + 1 = 3$. $g(1) = -3(1) + 1 = -2$. Sum = $3 + (-2) = 1$.

**33** **Answer: D** 18.0

$3x = 27 \Rightarrow x = 9$. Thus $2x = 18$.

**34** **Answer: C** 4

From the graph, line $f$ passes through $(-8, 2)$ and $(0, -2)$, yielding $f(x) = -0.5x - 2$. Parabola $g$ has roots at $-4$ and $0$ and vertex $(-2, 4)$, yielding $g(x) = -x(x + 4) = -x^2 - 4x$. $f(-6) = 1$ and $g(-1) = 3$. Sum is 4.

**35** **Answer: A** -1.5

$m = (-2 - 4)/(1 - (-3)) = -6/4 = -1.5$.

**36** **Answer: C** 57 g

$0.05674$ kg $= 56.74$ g. Rounded to the nearest gram is 57 g.

**37** **Answer: B** -3

Discriminant $\Delta = b^2 - 4ac = (-3)^2 - 4(3)(1) = 9 - 12 = -3$.

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**38** **Answer: C** $2/3$

$(5^2)^{3x-1} = (5^3)^x \Rightarrow 5^{6x-2} = 5^{3x} \Rightarrow 6x - 2 = 3x \Rightarrow 3x = 2 \Rightarrow x = 2/3.$

**39** **Answer: A** $y = -2x + 2$

The graphed line passes through $(0, 2)$ and $(-4, 0)$, so its slope is $(2 - 0)/(0 - -4) = 1/2$. A perpendicular line has slope $-2$. Equation: $y - 0 = -2(x - 1) \Rightarrow y = -2x + 2$.

**40** **Answer: C** (-1, 3)

Vertex x-coordinate $= -b/(2a) = -(-4)/(2 \times -2) = 4/-4 = -1$. $y = f(-1) = -2(1) + 4 + 1 = 3$. The point is $(-1, 3)$.

**41** **Answer: B** -4

$|2(-1) - 3(-2)| + 4(-2) = |-2 + 6| - 8 = |4| - 8 = -4.$

**42** **Answer: C** 4

From the graph, line $f$ passes through $(-8, 2)$ and $(0, -2)$, yielding $f(x) = -0.5x - 2$. Parabola $g$ has roots at $-4$ and $0$ and vertex $(-2, 4)$, yielding $g(x) = -x(x + 4) = -x^2 - 4x$. $f(-6) = 1$ and $g(-1) = 3$. Sum is $4$.

**43** **Answer: D**

See the answer key.

**44** **Answer: A** $-7 + 3i$

$(2i + 5)(i - 1) = 2i^2 - 2i + 5i - 5 = 2(-1) + 3i - 5 = -7 + 3i.$

**45** **Answer: A** $y = \dfrac{4x^2 - 2x - 1}{3x}$

$3yx = 4x^2 - 2x - 1 \Rightarrow y = \frac{4x^2-2x-1}{3x}$.

**46** **Answer: A** 1

From the graph, line $f$ passes through $(-8, 2)$ and $(0, -2)$, yielding $f(x) = -0.5x - 2$. Parabola $g$ has roots at $-4$ and $0$ and vertex $(-2, 4)$, yielding $g(x) = -x(x + 4) = -x^2 - 4x$. $f(-6) = 1$ and $g(-1) = 3$. Sum is $4$.

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**47** **Answer: C** 150.0 cm^2

In right triangle ACD: $CD^2 = 15^2 - 9^2 = 144 \Rightarrow CD = 12$. In right triangle BCD: $BD^2 = 20^2 - 12^2 = 256 \Rightarrow BD = 16$. Base $AB = 9 + 16 = 25$. Area = $(1/2) * 25 * 12 = 150$.

**48** **Answer: C** 12.5

$3x = 27 \Rightarrow x = 9$. Thus $2x = 18$.

**49** **Answer: A** 1

By the Remainder Theorem, $P(-1) = 0$. $3(-1)^3 - 2(-1) + m = -3 + 2 + m = -1 + m = 0$. Thus $m = 1$.

**50** **Answer: B** 2

$3x < 9 \Rightarrow x < 3$. The positive integers are 1 and 2. Their product is $1 \times 2 = 2$.

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